

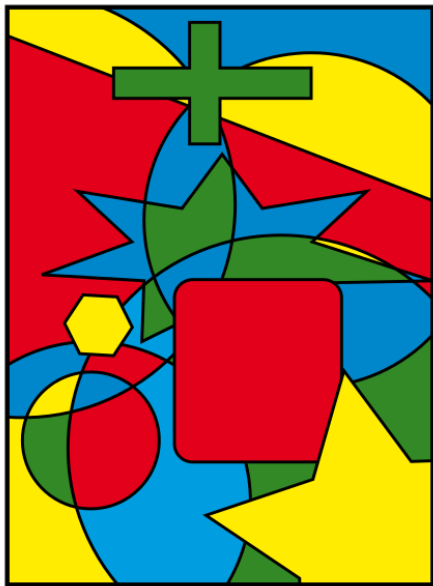
10-planarity-coloring: 平面图与图着色 (Planarity and Coloring)

魏恒峰

hfwei@hnu.edu.cn

2026 年 05 月 21 日



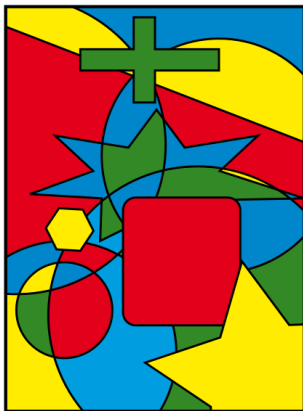


Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent regions** share the same color.*

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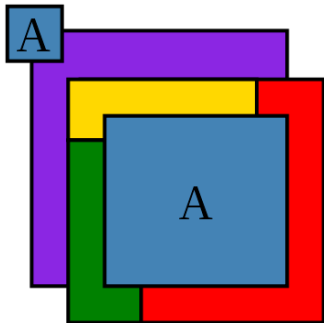


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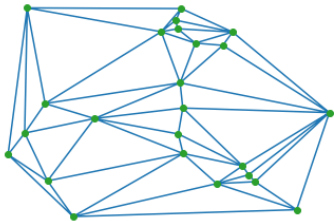
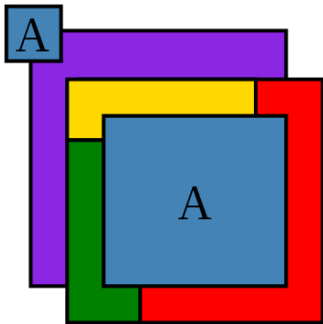
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Regions should be **contiguous**.

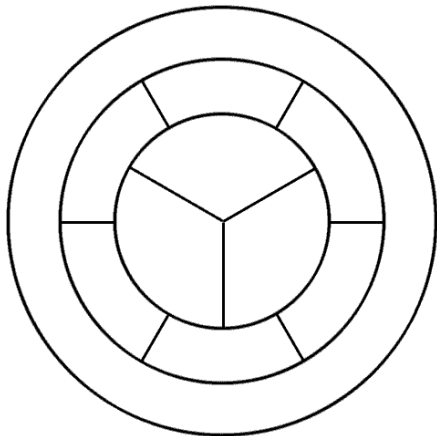
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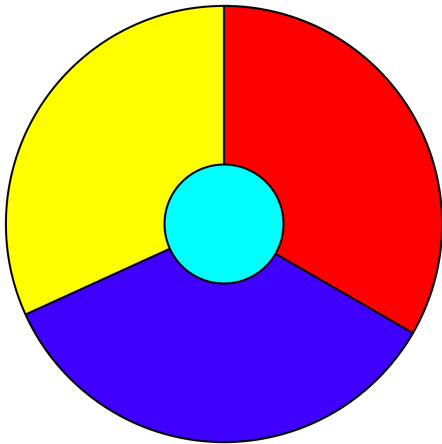
Adjacent regions share a segment.

Regions should be **contiguous**.



4 colors are necessary

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Every region is adjacent to the other 3 regions.

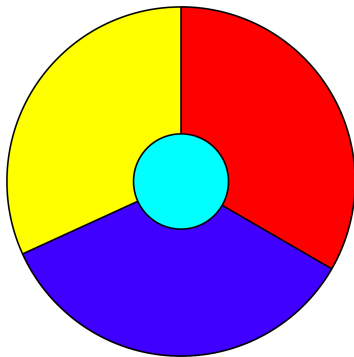
What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

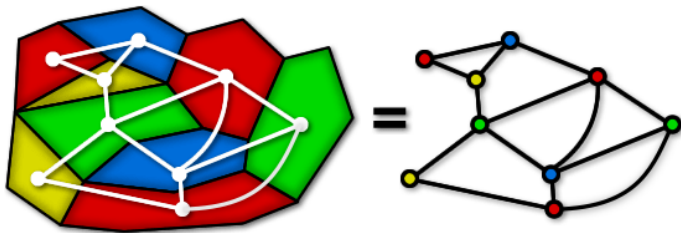
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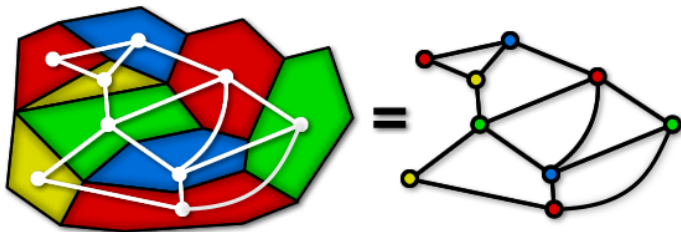


IMPOSSIBLE™

What does Four Color Theorem to do with Graph Theory?



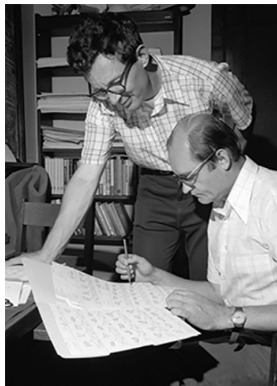




Every map produces a **planar** graph.

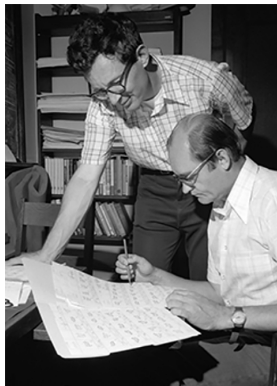
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I will *not* show its proof (which I don't understand either)!

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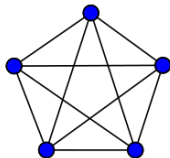


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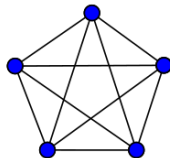


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IMPOSSIBLE



What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

Theorem

K_5 is not *planar*.

Theorem

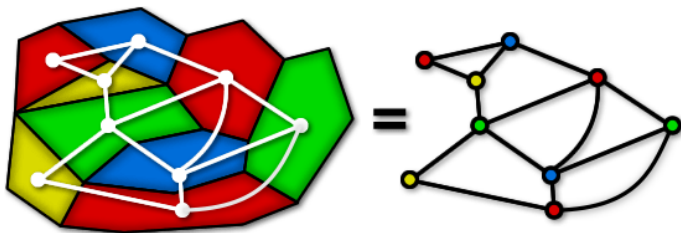
Every simple planar graph is 6-colorable.

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Theorem (Percy John Heawood (1890))

Every simple planar graph is 5-colorable.



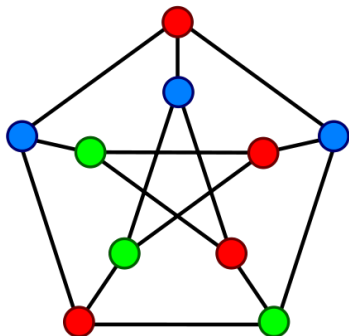
Graph Coloring Problem

Definition (k -Colorable (k -可着色的))

If G is a connected undirected graph without loops, then G is **k -colorable** if its vertices can be colored in k colors so that adjacent vertices have different colors.

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The Petersen graph is ≥ 3 -colorable.

Definition (k -Chromatic (k -色数的))

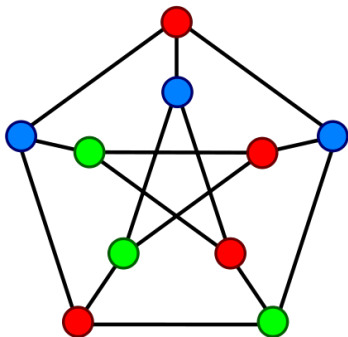
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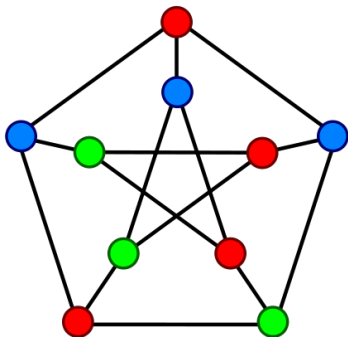


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The Petersen graph is **3**-chromatic.
(It contains an **odd** cycle.)

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A graph is 1-colorable iff

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A graph is 1-colorable iff it is the empty graph with 1 vertex.

Theorem

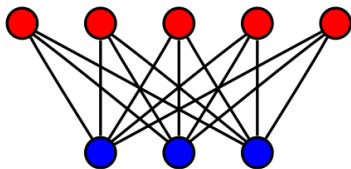
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Theorem

*A graph is 2-colorable iff it is **bipartite**.*

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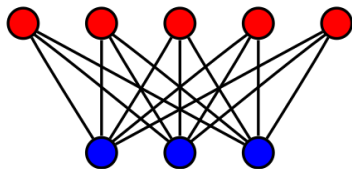
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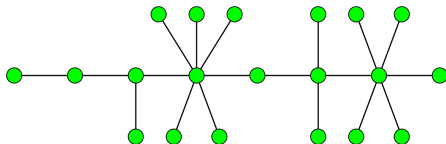
$K_{5,3}$

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$K_{5,3}$



Trees are bipartite.

Theorem (Characterization of Bipartite Graphs)

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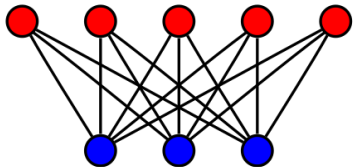
*A graph is 2-colorable iff it is **bipartite**.*

*A graph is **bipartite** iff it does not contain any **odd** cycles.*

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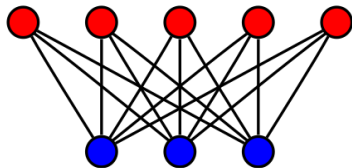


$K_{5,3}$

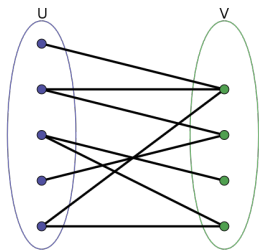
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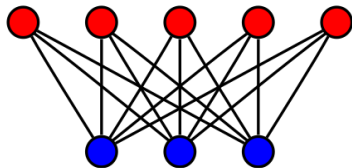
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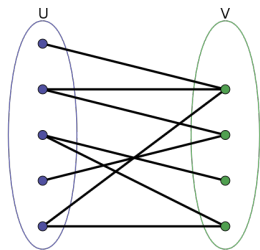
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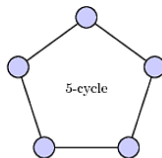
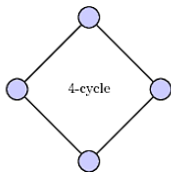
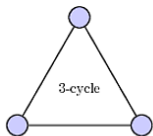


Lemma

Every tree is bipartite and is thus 2-colorable.

Lemma (Characterization of Bipartite Graphs ($\implies$))

If a graph is *bipartite*, then it does not contain any *odd* cycles.



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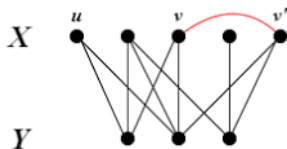
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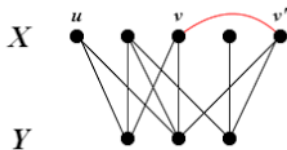
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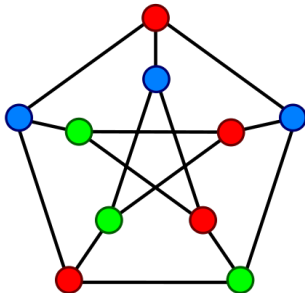
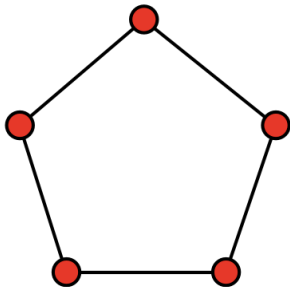
$u \rightsquigarrow v \rightarrow v' \rightsquigarrow u$ is an **odd** cycle.

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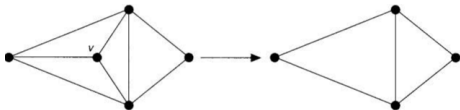
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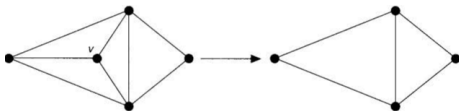
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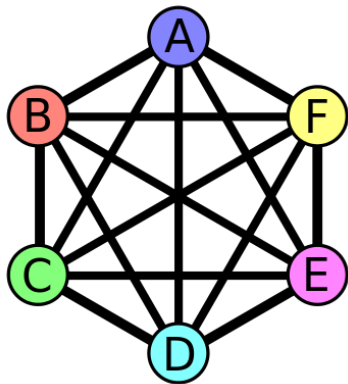
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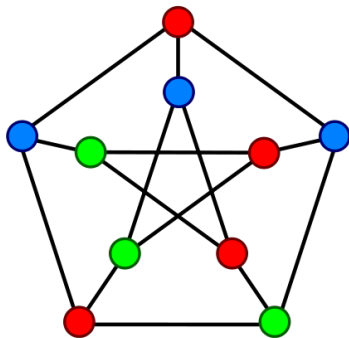
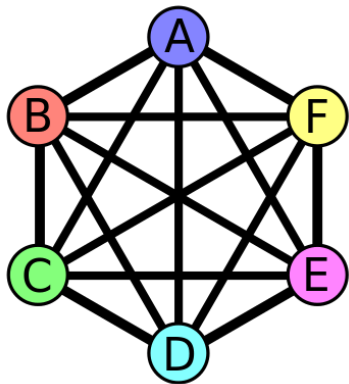
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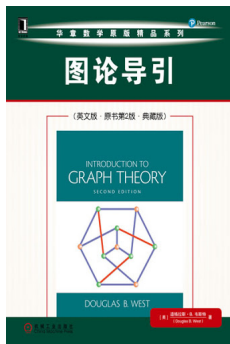
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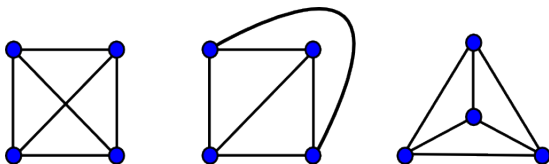
Theorem 5.1.22

Definition (Planar Graph (平面图))

A **planar graph** is a graph that **can** be drawn in the plane without **edge crossings**.

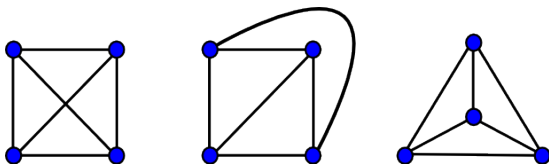
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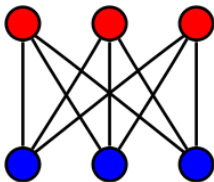


Theorem (K. Wagner (1936); I. Fáry (1948))

*Every simple planar graph can be drawn with **straight lines**.*

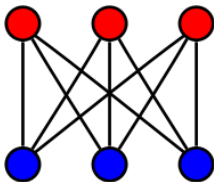
Theorem (Kazimierz Kuratowski, 1930)

The utility graph $K_{3,3}$ is non-planar.



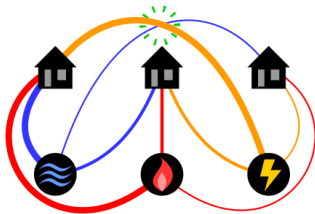
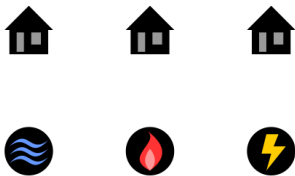
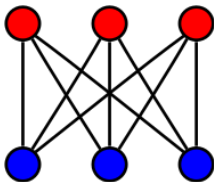
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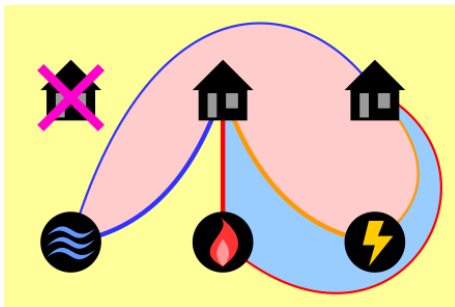
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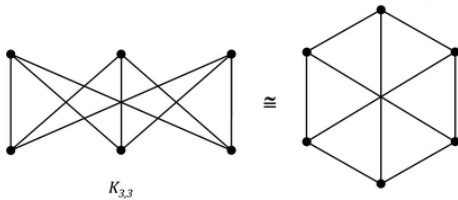
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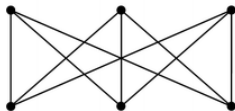
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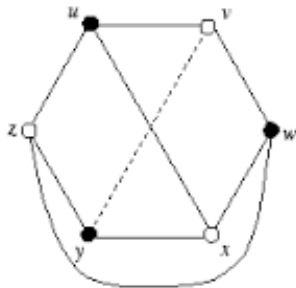
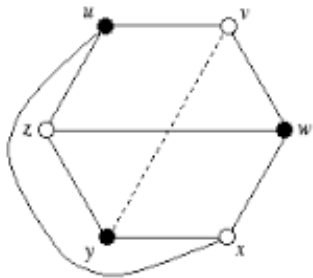
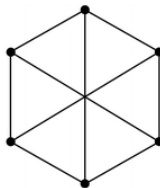
Proof without Words

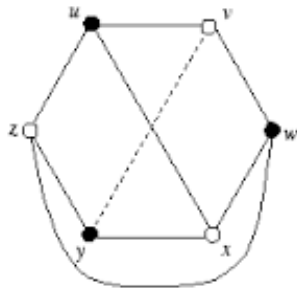
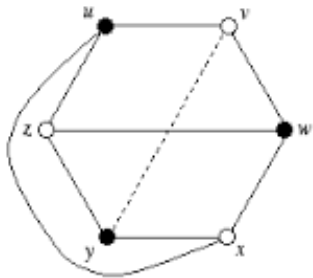
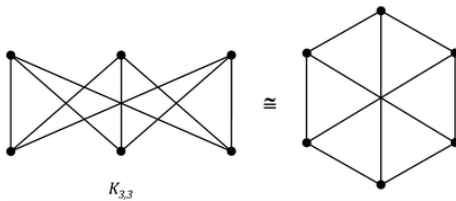




$K_{3,3}$

$\cong$

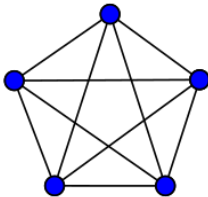




$$\text{cr}(K_{3,3}) = 1 \quad (\text{crossing number})$$

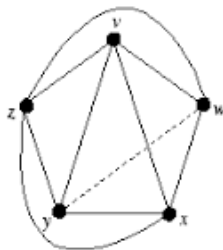
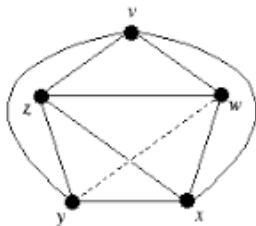
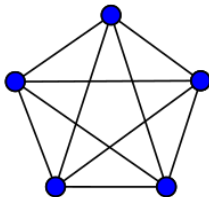
Theorem

K_5 is non-planar.



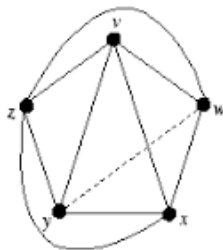
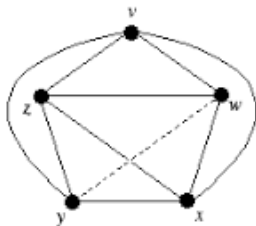
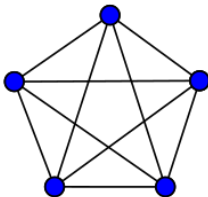
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$$\text{cr}(K_5) = 1$$

Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.

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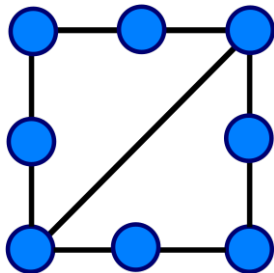
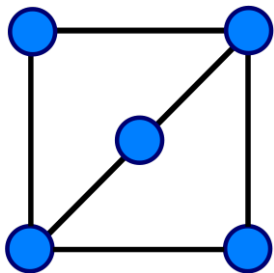
*“The K in K_5 stands for Kazimierz,
and the K in $K_{3,3}$ stands for Kuratowski.”*

Definition (Homeomorphic)

Two graphs are **homeomorphic** if one can be obtained from another by **inserting or contracting** vertices of **degree 2**.

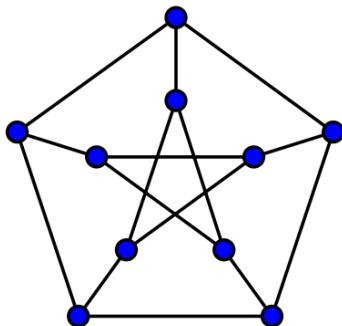
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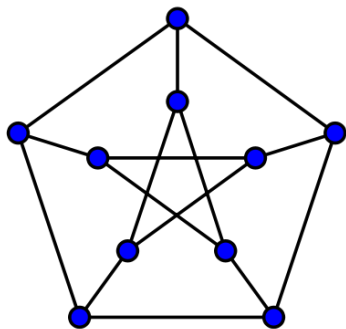


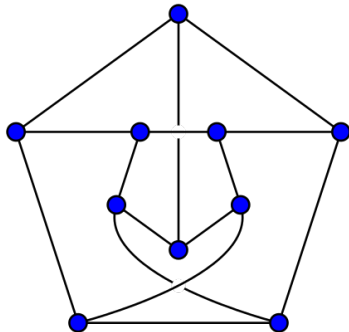
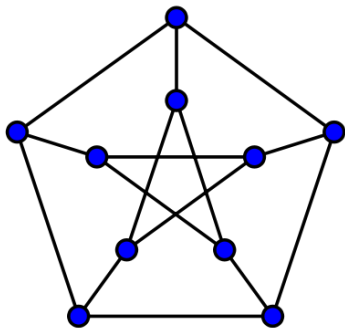
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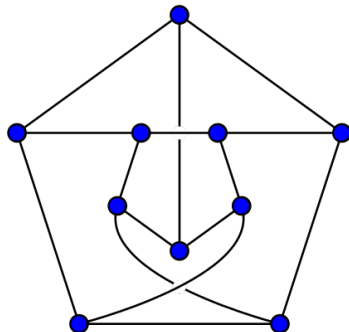
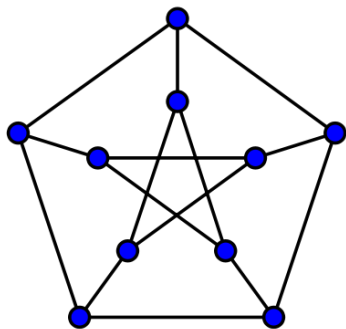
The Petersen graph is non-planar.



[https://github.com/courses-at-nju-by-hfwei/
discrete-math-lectures/blob/main/11-planarity-coloring/
figs/Kuratowski-Petersen.gif](https://github.com/courses-at-nju-by-hfwei/discrete-math-lectures/blob/main/11-planarity-coloring/figs/Kuratowski-Petersen.gif)







$$\text{cr}(\text{Petersen Graph}) = 2$$

A planar graph should not has too many edges.

Theorem (Euler's Formula, 1750)

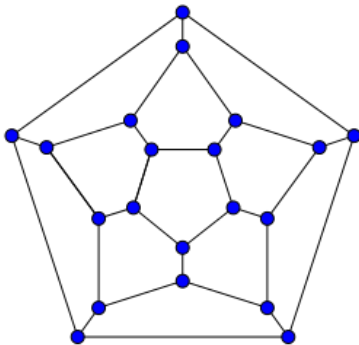
Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$

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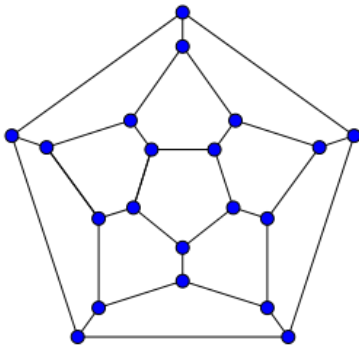
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$$n - m + f = 20 - 30 + 12 = 2$$

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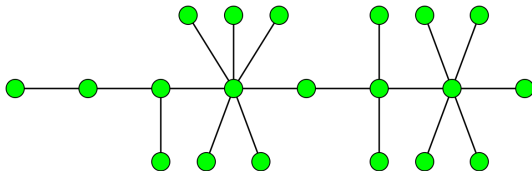
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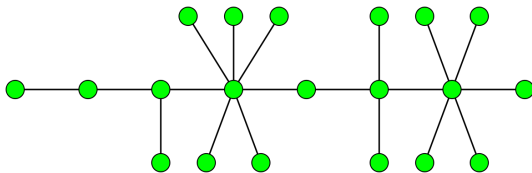
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$$n - m + f = n - (n - 1) + 1 = 2$$

By induction on the number of edges of G .

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If G is a tree, we are done.

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Let e be an edge in some cycle of G .

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Consider $G' = G - e$.

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Consider $G' = G - e$.

$$n - (m - 1) + (f - 1) = 2$$

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Let e be an edge in some cycle of G .

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Therefore,

$$n - m + f = 2$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

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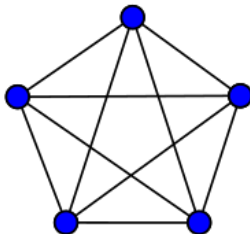
$$3f \leq 2m$$

Double Counting:

each face is bounded by ≥ 3 edges;
each edge bounds 2 faces

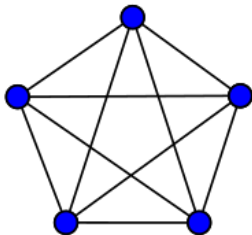
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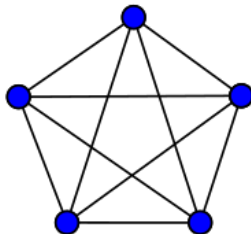
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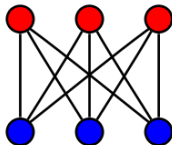


$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$

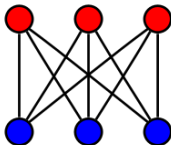
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$K_{3,3}$ is non-planar.



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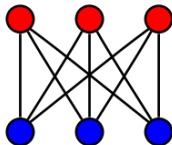
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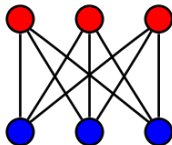


$$m \leq 3n - 6$$

$$9 \leq 3 \times 6 - 6$$

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FAILED

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Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. *If G has no triangles, then*

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Suppose that, **by contradiction**, $\delta(G) \geq 6$.

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Thus, G is 6-colorable.

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G contains a vertex v of degree ≤ 5 .

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Thus, G is 6-colorable. ($\deg(v) \leq 5$)

Theorem (Percy John Heawood (1890))

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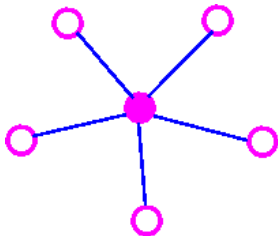
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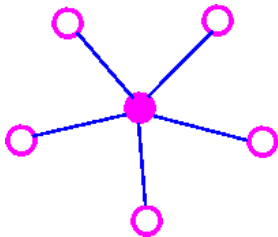
If $\deg(v) < 5$, G is 5-colorable.

Now assume that $\deg(v) = 5$.



$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

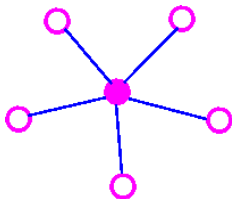
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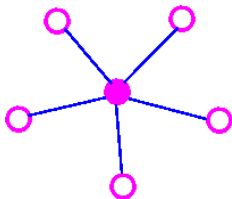
$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

If v_1, v_2, v_3, v_4 , and v_5 uses < 5 colors, we are done.

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.

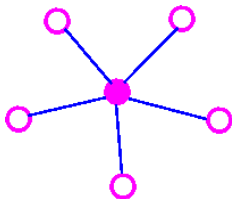


Now assume that v_1, v_2, v_3, v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

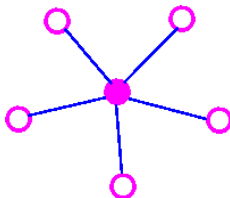
Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a red-blue path.

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.

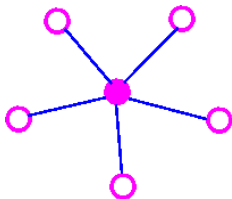


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$$v_1 \in S, \quad v_3 \notin S$$

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.



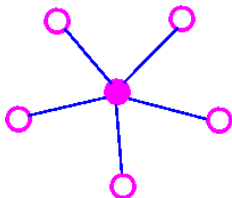
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Interchange the colors of the vertices in S

Now assume that v_1, v_2, v_3, v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
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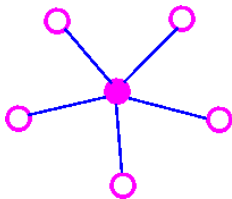
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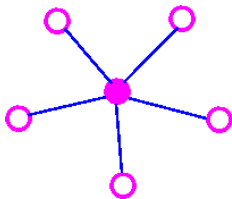
Interchange the colors of the vertices in S

Coloring v **red** produces a 5-coloring of G .

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

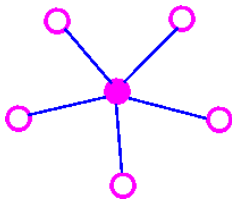


Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
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There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

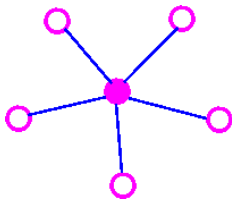
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There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

(Otherwise, G' and thus G is non-planar.)

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

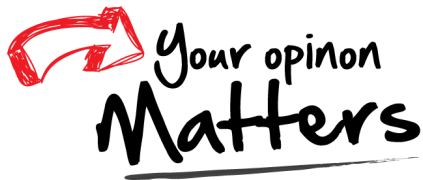


There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

(Otherwise, G' and thus G is non-planar.)

By similar argument, G is 5-colorable.

Thank
You!



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